CLINICAL PROGRAMMING BOOTCAMP • MODULE 04

Reading Raw Clinical Data

Before you can map anything, you have to get the data in — with the right types, the right dates, and your eyes open for what's wrong with it.

Hands-on: Notebook 03 · Importing Raw Data (SAS & R)

MODULE GOALS

By the end of this module you can…

1. **Import a raw CSV correctly** in both SAS and R, with the types you intended.
2. **Recognise the four data types** character, numeric, date, and missing — and tell them apart.
3. **Spot the classic traps** leading zeros, mixed date formats, blanks, wide structures.
4. **Inspect before you map** run a standard checklist on any unfamiliar dataset.
5. **Plan a mapping** list exactly what must change to reach SDTM.

THE STARTING POINT

What a raw extract actually looks like

One file per CRF form — note Demographics and Disposition are separate forms, completed at different times.

File	CRF form	Structure	Rows
`dm_raw.csv`	**Demographics (screening)**	one row per subject	8
`ds_raw.csv`	**Disposition / end of study**	one row per subject	8
`ex_raw.csv`	**Study drug exposure**	one row per subject	8
`ae_raw.csv`	**Adverse events**	one row per event	10
`cm_raw.csv`	**Concomitant meds**	one row per medication	8
`vs_raw.csv`	**Vital signs**	WIDE — one row per visit	24
`lb_raw.csv`	**Laboratory**	tall — one row per test	48

Note vs_raw.csv is WIDE and lb_raw.csv is TALL — the same clinical idea, two different shapes. You'll handle both.

Everything is one of three types (plus missing)

Character

Text. Anything you don't do arithmetic on.

```
"ABC-01"  "F"  "bad headache"
IDs, codes, names, ISO dates
```

Numeric

Numbers you can calculate with.

```
50   36.7   122
Doses, results, ages
```

Date

A calendar point. Stored as a number internally, but SDTM wants ISO text.

```
2024-03-01
(character in SDTM!)
```

The fourth state: MISSING. A value that was not collected. In SAS a missing number is `.` and missing text is `""`; in R both are `NA`. Missing is never the same as zero or an empty answer.

TRAP 1

Type guessing can destroy IDs

Readers guess column types from the values. An ID made only of digits looks like a number — and different tools make different guesses on the SAME file.

In the file	SAS PROC IMPORT base R read.csv()	readr read_csv()
SITEID = 01 SUBJID = 001	SITEID = 1 SUBJID = 1	SITEID = "01" SUBJID = "001"

→ If the zeros are lost, USUBJID becomes ABC-01-1-1 instead of ABC-01-01-001 — wrong for every subject, in every domain.

SAS

```
/* control the types yourself */
data dm_raw;
  infile "dm_raw.csv" dsd firstobs=2;
  length siteid $3 subjid $5;
  input studyid $ siteid $ subjid $ ...;
run;
```

R

```
# declare intent, never rely on luck
read_csv("dm_raw.csv",
  col_types = cols(
    SITEID = col_character(),
    SUBJID = col_character(),
    .default = col_guess()))
```

TRAP 2

No raw date is ISO — convert every one

`14-MAY-1969`	DD-MMM-YYYY — the CRF standard	*found in dm, ds, ex, vs, lb*
`15/03/2024`	DD/MM/YYYY	*found in ae, cm*
`10-Mar-2024`	DD-Mon-YYYY	*found in ae, cm*

Three dangers. **(1)** 01/03/2024 is ambiguous — 1 March or 3 January? Check the data dictionary, never guess. **(2)** A blank date means ongoing or not done — leave it null, never substitute today's date. **(3)** DD-MMM-YYYY text does NOT sort chronologically — convert before any min/max.

TRAP 3

Blank does not mean zero

1	**Not collected** The test wasn't done at this visit.	HEIGHT is blank at BASELINE and WEEK 4 — height is measured once, at screening.
2	**Still ongoing** The event or medication hasn't ended.	AEENDT is blank for the Nausea event — it was still ongoing at the end of the study.
3	**Genuinely unknown** It was asked but the answer isn't known.	ETHNIC = 'Unknown' for one subject — that is a real recorded value, not a blank.

Different reasons, same appearance in the file. Understanding which one you're looking at is a clinical question, not a technical one.

TRAP 4

Wide and tall are both “normal”

WIDE — vs_raw.csv

SUBJID	VISIT	SYSBP	DIABP	PULSE
001	SCREENING	122	80	68
001	BASELINE	120	78	66

One row per visit · one column per measurement

TALL — lb_raw.csv

SUBJID	VISIT	LBTEST	LBORRES
001	BASELINE	Hemoglobin	14.2
001	BASELINE	Hematocrit	42

One row per measurement · a test-name column

SDTM Findings domains are always TALL — so wide data must be transposed on the way in.

Recognising the shape is the first decision you make about a Findings file: does it need transposing, or is it already in the right form?

`VS needs transposing (24 rows → 128).` `LB does not (48 rows → 48).`

TRAP 5

The same idea, spelled many ways

Coded numbers	`SEX = 1 / 2`	You need the dictionary to know 1=Male, 2=Female
Inconsistent case	`White / white / 'White '`	Trim and upper-case before matching
Free text	`'bad headache' / 'Headache'`	Requires dictionary coding (MedDRA) — not a simple lookup
Synonyms	`'No' / 'N'`	Both mean the same; map both to CT 'N'

Cleaning these is Controlled Terminology work — Module 08. For now, just learn to spot them.

DISCIPLINE

The inspection checklist

Run this on every unfamiliar dataset, every time. It takes two minutes and saves hours.

1 **How many rows and columns?**

Does the row count match what you expected?

2 **What type is every column?**

Especially the IDs — check the leading zeros survived.

3 **How many distinct subjects?**

Compare against DM. Any subject not in DM is a problem.

4 **Which columns have blanks?**

And what does each blank mean?

5 **What are the distinct values?**

For any column you'll map to CT — look at every value.

6 **What date formats appear?**

Check for more than one format in the same column.

THE OUTPUT OF INSPECTION

Gap analysis: ABC-01 raw → SDTM

Inspection produces a to-do list. Here is the real one for our study.

What's wrong in the raw data	What you must do	Affects
SUBJID not unique study-wide	**Build USUBJID**	`all`
SEX stored as 1 / 2	**Apply CT → M / F**	`DM`
RACE free text, mixed case	**Trim, upper-case, map to CT**	`DM`
Birth date, no age	**Derive AGE and AGEU**	`DM`
Dates in 3 formats	**Convert all to ISO 8601**	`AE, CM`
No study day anywhere	**Derive --DY from RFSTDTC**	`all`
No record sequence	**Derive --SEQ**	`all`
VS is wide	**Transpose to --TESTCD / --TEST**	`VS`

Every one of these is documented in data/mapping_specification.md — your reference for the rest of the bootcamp.

AVOID THESE

Import mistakes that cost hours

Not checking the row count
If you imported 23 rows from a 24-row file, something was silently dropped.

Letting IDs become numbers
Leading zeros vanish and your USUBJID is wrong for every subject.

Assuming one date format
Check every date column for more than one pattern.

Treating blanks as zero
A missing weight is not a weight of 0 kg.

Editing the raw file
Never. Fix it in code so the step is documented and repeatable.

Skipping the dictionary
The data dictionary tells you what 1/2 and DD/MM mean. Read it first.

WHAT'S NEXT

Get the data in, then build your first domain

1. **Notebook 03** Importing Raw Data — read all six files, run the checklist (SAS & R)
2. **Module 05** Building the DM domain — Special Purpose, one row per subject
3. **Notebook 04** Build DM end to end and check it against the reference dataset

You now have the finished answer in data/sdtm/ — but build it yourself first, then compare.